

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		A {disease / it} that may be {passed / spread / transmitted / transferred} from {one individual to another / organism to organism / between individuals} Ignore any reference to mode of transmission	1			1		
		(ii)		Uses {weakened / killed / attenuated} pathogen / antigens / toxins} to stimulate {the immune response / production of antibodies / production of memory cells}	1			1		
		(iii)		(As transmitted through) {coughing and sneezing / droplet infection / airborne / aerosol} (1) overcrowded conditions / high population density areas / {high levels / increased risk} of contact (with infected people) (1)		1	1	2		
	(b)	(i)		A description of graph antibody levels constant / zero + then increase + then decrease (1) Explanation – any three (x1) from: B latent period / {macrophages / APC} carry out phagocytosis / antigen presentation takes place (1) C clonal selection / activation of B cells / T helpers release cytokines to stimulate B cells / B cells undergo clonal {expansion / proliferation} (1) D (increases as) plasma cells release antibodies (1) E (decreases as) antibodies are broken down (1)	2	2		4		

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		(ii)		Any four (x1) from A. {Short / no} latent period (1) B. due to memory cells (1) C. <u>rapid</u> production of plasma cells / memory cells undergo clonal expansion faster (1) D. Antibody concentrations increase {faster / sooner / to higher concentrations} (than primary response) (1) E. Antibody concentrations stay higher for longer / remain in the bloodstream for longer (1)	2	2		4												
	(c)	(i)		90.9/ 90.91/91 = 2 marks Award 1 mark for sight of 132-12/132 x 100		2		2	2											
		(ii)		<table border="1"><tr><td></td><td>Advantage</td><td>Disadvantage</td></tr><tr><td>Drug A and antibiotics</td><td>TB eradicated / no dormant TB remaining / M. tuberculosis killed (1)</td><td>antibiotics needed / (re)activating {TB / M. tuberculosis} could cause TB (1)</td></tr><tr><td>Drug B alone</td><td>antibiotics not needed / {TB / M. tuberculosis} not reactivated / M. tuberculosis remains dormant (1)</td><td>take drug B for life / (dormant) M. tuberculosis {still present / not killed} (1) NOT: symptoms reappear if drug removed</td></tr></table>		Advantage	Disadvantage	Drug A and antibiotics	TB eradicated / no dormant TB remaining / M. tuberculosis killed (1)	antibiotics needed / (re)activating {TB / M. tuberculosis} could cause TB (1)	Drug B alone	antibiotics not needed / {TB / M. tuberculosis} not reactivated / M. tuberculosis remains dormant (1)	take drug B for life / (dormant) M. tuberculosis {still present / not killed} (1) NOT: symptoms reappear if drug removed				4	4		3
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	(d)			Any two (x1) from: <u>Plasmodium</u> not affected by antibiotics (1) Drugs not effective whilst <u>plasmodium</u> in RBC (1) Resistance to {drugs / named treatments} (1) Reference to antigenic variation / high mutation rate (1)		2		2												
				Question 7 total	6	9	5	20	2	3										